

Definition of the differential: If $\frac{dy}{dx} = f'(x)$ then $dy = dx \cdot f'(x)$

Fundamental Theorem of Calculus:

$$\int_a^b f(x)dx = F(b) - F(a)$$

1) Find the indefinite integral $\int (3 + 7x) \cdot dx$

Let $u = 3 + 7x$

a) $\frac{du}{dx} = \underline{\hspace{1cm}}?$

b) By definition of Differential: $du = \underline{\hspace{1cm}}?$

c) $\frac{1}{7} \cdot du = \underline{\hspace{1cm}}?$

d) Write the integral $\int (3 + 7x) \cdot dx$ in terms of u and then evaluate.

$$\int (3 + 7x) \cdot dx = \int \underline{\hspace{1cm}}? \cdot du = \underline{\hspace{1cm}}?$$

e) $\int (3 + 7x) \cdot dx = \underline{\hspace{1cm}}?$ (Note: Answer must be in terms of x .)

2) Find the indefinite integral $\int \sqrt{16 - x^2} \cdot x \cdot dx$

Let $u = 16 - x^2$

a) $\frac{du}{dx} = \underline{\hspace{1cm}} ?$

b) By definition of Differential: $du = \underline{\hspace{1cm}} ?$

c) $\frac{-1}{2} \cdot du = \underline{\hspace{1cm}} ?$

d) Write the integral $\int \sqrt{16 - x^2} \cdot x \cdot dx$ in terms of u and then evaluate.

$$\int \sqrt{16 - x^2} \cdot x \cdot dx = \int \underline{\hspace{1cm}} ? \cdot du = \underline{\hspace{1cm}} ?$$

e) $\int \sqrt{16 - x^2} \cdot x \cdot dx = \underline{\hspace{1cm}} ?$ (Note: Answer must be in terms of x .)

3) Find the indefinite integral $\int (x^2 + 3) \cdot x \cdot dx$

Let $u = x^2 + 3$

a) $\frac{du}{dx} = \underline{\quad ? \quad}$

b) By definition of Differential: $du = \underline{\quad ? \quad}$

c) $\frac{1}{2} \cdot du = \underline{\quad ? \quad}$

d) Write the integral $\int (x^2 + 3) \cdot x \cdot dx$ in terms of u and then evaluate.

$$\int (x^2 + 3) \cdot x \cdot dx = \int \underline{\quad ? \quad} \cdot du = \underline{\quad ? \quad}$$

e) $\int (x^2 + 3) \cdot x \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x .)

4) Find the indefinite integral $\int (x^3 - 5) \cdot x^2 \cdot dx$

Let $u = x^3 - 5$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) By definition of Differential: $du = \underline{\hspace{2cm}}?$

c) $\frac{1}{3} \cdot du = \underline{\hspace{2cm}}?$

d) Write the integral $\int (x^3 - 5) \cdot x^2 \cdot dx$ in terms of u and then evaluate.

$$\int (x^3 - 5) \cdot x^2 \cdot dx = \int \underline{\hspace{2cm}}? \cdot du = \underline{\hspace{2cm}}?$$

e) $\int (x^3 - 5) \cdot x^2 \cdot dx = \underline{\hspace{2cm}}?$ (Note: Answer must be in terms of x .)

5) Find the indefinite integral $\int (1-x^2)^{1/4} \cdot x \cdot dx$

Let $u = 1 - x^2$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) By definition of Differential: $du = \underline{\hspace{2cm}}?$

c) $\frac{-1}{2} \cdot du = \underline{\hspace{2cm}}?$

d) Write the integral $\int (1-x^2)^{1/4} \cdot x \cdot dx$ in terms of u and then evaluate.

$$\int (1-x^2)^{1/4} \cdot x \cdot dx = \int \underline{\hspace{2cm}}? \cdot du = \underline{\hspace{2cm}}?$$

e) $\int (1-x^2)^{1/4} \cdot x \cdot dx = \underline{\hspace{2cm}}?$ (Note: Answer must be in terms of x .)

6) Find the indefinite integral $\int \frac{4x^2}{(1+x^3)^2} \cdot dx$

Hint: $\int \frac{4x^2}{(1+x^3)^2} \cdot dx = 4 \cdot \int \frac{x^2}{(1+x^3)^2} \cdot dx = 4 \cdot \int \frac{1}{(1+x^3)^2} \cdot x^2 \cdot dx$

Let $u = 1 + x^3$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) By definition of Differential: $du = \underline{\hspace{2cm}}?$

c) $\frac{1}{3} \cdot du = \underline{\hspace{2cm}}?$

d) Write the integral $4 \cdot \int \frac{1}{(1+x^3)^2} \cdot x^2 \cdot dx$ in terms of u and then evaluate.

$$4 \cdot \int \frac{1}{(1+x^3)^2} \cdot x^2 \cdot dx = \int \underline{\hspace{2cm}}? \cdot du = \underline{\hspace{2cm}}?$$

e) $\int \frac{4x^2}{(1+x^3)^2} \cdot dx = 4 \cdot \int \frac{1}{(1+x^3)^2} \cdot x^2 \cdot dx = \underline{\hspace{2cm}}?$ (Note: Answer must be in terms of x .)

7) Find the indefinite integral $\int \left(4 + \frac{1}{x}\right) \cdot \frac{1}{x^2} \cdot dx$

Let $u = 4 + \frac{1}{x} = 4 + x^{-1}$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}$

b) By definition of Differential: $du = \underline{\hspace{2cm}}$

c) $-1 \cdot du = \underline{\hspace{2cm}}$

d) Write the integral $\int \left(4 + \frac{1}{x}\right) \cdot \frac{1}{x^2} \cdot dx$ in terms of u and then evaluate.

$$\int \left(4 + \frac{1}{x}\right) \cdot \frac{1}{x^2} \cdot dx = \int \underline{\hspace{2cm}} \cdot du = \underline{\hspace{2cm}}$$

e) $\int \left(4 + \frac{1}{x}\right) \cdot \frac{1}{x^2} \cdot dx = \underline{\hspace{2cm}}$ (Note: Answer must be in terms of x .)

